

Working memory

Hypothesis:

We are interested in creating a model that can solve an array of delay-match-to-sample (DMTS) tasks of escalating complexity that requires the model to maintain an image(s) in working memory and compare it with a later image under delays and transformations to determine whether they are repeated. I posit that adding a recurrent connectivity layer immediately before output on a standard pre-trained CNN (e.g. ResNet-50) will be sufficient to perform well on these tasks, and will test this model among an assortment of others specified below.

Literature review:

Working memory is the temporary storage and manipulation of information, though the scope of the topic has expanded dramatically since its inception. From the hypothesis, we are interested in evaluating models' working memory, and identifying their working memory capacity.

Developing models that exhibit working memory, and evaluating their biological plausibility, has been addressed before in a variety of standard models. Recurrent neural networks (RNNs) have become popular models for evaluating cognitive theories due to their ease of training, but standard RNNs are not designed to complete visual tasks, such as identifying repeated images; similarly convolutional neural networks (CNNs), while used for visual data, struggle in tasks where neural representations have longer temporal dependencies. Thus we consider introducing recurrent connectivity layers to standard CNN models, letting neurons feed their outputs back into their own inputs to create internal memory storage, potentially allowing the modified CNN to learn temporal dependencies on image tasks.

For the tasks at hand, tasks 1, 2, and 4 have been benchmarked in humans, and they are standard to the field of working memory (I elaborate on this in the "Project overview" section).

While I initially considered reinforcement learning (RL) actor-critic models as a possible testable model, and though RL models have been designed for working memory analysis, I have found that RL models will likely not be fruitful given the nature of the DMTS tasks we test (discussion of this is in the "Preliminary progress" section).

Citations:

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 Chelazzi, L., Miller, E. K., Duncan, J. & Desimone, R. (2001). Responses of neurons in macaque area V4 during memory-guided visual search. Cerebral Cortex 11, 761-772.

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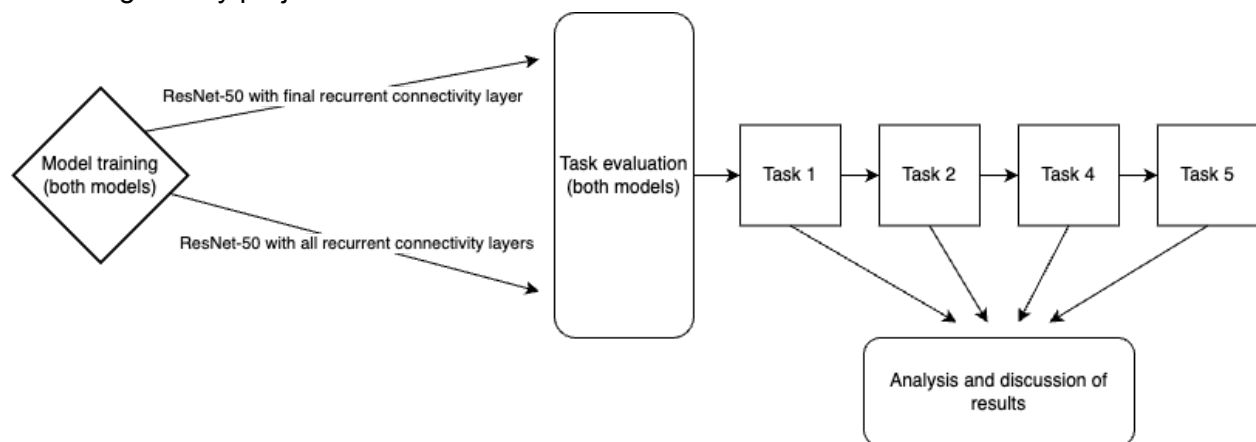
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Wilhelm, O., Hildebrandt, A., & Oberauer, K. (2013). What is working memory capacity, and how can we measure it? *Frontiers in Psychology*, 4, 433–433. <https://doi.org/10.3389/fpsyg.2013.00433>

Working memory: theories, models, and controversies. Baddeley A. Annu Rev Psychol. 2012;63:1-29. doi: 10.1146/annurev-psych-120710-100422. Epub 2011 Sep 27.

Project overview:

The design of my project is as follows:



The models I will train and test on have architectures of recurrent connectivity immediately before output on ResNet-50, and recurrent connectivity on all layers on ResNet-50.

I will begin by training these models. I will be using the ImageNet dataset, which is publicly available and free for non-commercial use (this dataset seems pretty well-known; it constitutes approx. 14 million images, in RGB format, with various resolutions, each with corresponding WordNet ID). Models will be trained on a small subset of ImageNet (I intend to train and test on a subset of 1,000 “synsets”, i.e. different image concepts, of the >100,000 in ImageNet, as given

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in ILSVRC2010, here: <https://image-net.org/challenges/LSVRC/2010/#data>). These images will be cropped to 224 x 224 patches.

In task evaluation, the models will be sequentially fed images from ImageNet, which are possibly transformed (e.g. subject to rotation, scaling, position changes to test invariance - I intend to draw uniformly at random for degree of change) - I intend to run three versions of each task: one where images are not transformed; one where repeated images are subject to scale augmentation, achieved through the PyTorch `Resize` function (that is then cropped to resolution 224 x 224 px using the `RandomCrop` function, fitting what ResNet-50 is typically trained on); and one where repeated images are subject to scale and horizontal flip, with the PyTorch `Resize` and `RandomHorizontalFlip` transformations. These transformations are made uniformly at random.

The tasks, ascending in complexity, are as follows:

- Task 1: We have a sequence of images where a sample image is shown, a delay is presented (i.e. empty images, with the number of images Δt randomly drawn within a set interval $[t_{min}, t_{max}]$ for each trial), and a test image is shown that is possibly a repeat of the sample image (potentially subject to transformation, as described above). Model output is $\{0, 1\}$ binary, indicating whether the test image is repeated; the accuracy is the percentage of times the model correctly determines whether the test image is repeated.
- Task 2: We inherit the task conditions from above, except now the delay includes $0 < n < \Delta t$ distractor images. Model output is $\{0, 1\}$ binary indicating whether the test image is repeated, and accuracy is again the percentage of times the model correctly determines whether the test image is repeated.
- Task 4: We inherit the conditions from prior tasks, except now the sample and test images can appear at any position in the sequence of images presented, and the sequences of images have arbitrary length. Model output is $\{0, 1\}$ binary indicating whether the test image is repeated, and accuracy is again the percentage of times the model correctly determines whether the test image is repeated.
- Task 5: We inherit and generalize the conditions from prior tasks, now with an arbitrary number of repetitions and positions. After each image the model outputs a $\{0, 1\}$ binary, indicating whether the test image is repeated; accuracy is the total percentage of times the model correctly determines whether the images are repeated. As this task is very general, (as with prior tasks) we can map accuracy as a function of the number of repetitions, positions, etc.

These tasks are as specified in the “models of working memory” page, here:

https://docs.google.com/document/d/1oV3Asf-8qEqA96_QWvBH-qOZuYJhHp-u32Rpe_GpE5E/edit?usp=sharing Half of trials will have a repetition, while half will not.

These styles of tasks, while interesting when applied to the visual realm, are standard to the analysis of working memory: task 1 is a textbook delayed match to sample task; (generalizing)

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with the incorporation of distractors, task 2 includes aspects of the dual task paradigm; (generalizing further) task 5 borrows elements of complex span tasks in storing information from many images in parallel. The metric for task evaluation, an accuracy score for correctly labelling an image as repeated or not, is standard. I will build off of much of the codebase given for training CNNs on ImageNet in Krizhevsky et al. for training and data augmentation; while I am not evaluating classification (rather, whether an image is repeated), much of the code is the same.

I will calculate the accuracy by which models identify repeated images as a function of whether the repeated images are transformed, how many images are presented before a repeated image can be evaluated, and (for task 5) how many repetitions are encountered. Of the tasks we use, tasks 1, 2, and 4 have been benchmarked in humans (Chelazzi et al. 1998), and tasks 1 and 2 have been benchmarked in monkeys (Chelazzi et al. 2001).

These benchmarks lay the groundwork for my analysis. I would note whether model performance is above chance and above animal performance, as well as how models perform relative to one another. I would also note how transformations, length before repeat, etc. influence accuracy. Beyond noting model performance, I hope to compare performance with that of humans and monkeys, and consider what behavior and memory capabilities of the concerned models are biologically plausible. This can allow discussion of why the model architecture may have similar representations/structure to animals. As my project is more exploratory, I find any tractable inference to be a success; for instance, were the performance of my models to replicate humans in particular conditions, I could ask what about model architecture (i.e. the introduction of recurrent connectivity) mimics behavioral data, and why the data at hand, and how far this relationship extends.

Key papers:

Krizhevsky et al. provides the codebase for much of my project, as well as laying the groundwork for training RNNs on ImageNet, and introducing standards for data augmentation of ImageNet.

Chelazzi et al., 1998 & 2001 provide human (for tasks 1, 2, and 4) and monkey (tasks 1 and 2) performance benchmarks for the tasks at hand, which underpins my analysis of the biological plausibility and comparison of behavior with the tested models.

Challenges:

The process of running many tasks on many models will be computational intensive. For this purpose, I have access to a SEAS cluster, and will otherwise run on Google Colab using a GPU. I am only training on a small portion of the ImageNet dataset. Additionally, I will restrict to running only a few modifications of each task, as specified above.

Preliminary progress:

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I have loaded the ImageNet dataset and have become familiar with the labelling system. Following this, I have trained a standard ResNet-50 on the specified subset of ImageNet, which simply hopes to classify the concept category of the image. I have also created equivalent ImageNet datasets subject to these transforms, and verified that the ResNet-50 can be trained and tested on the transformed dataset as well (to verify labels, etc. were correctly conserved). I am currently working on adding recurrent connectivity layers to ResNet-50, and the immediate next step is to train the two altered ResNet-50 architectures that I propose, and then initiate task evaluation.

I have additionally conducted literature review, and given my results have chosen to alter the scope of the project's tasks and model selection somewhat. I have learned that (stimulated by discussion in office hours), while reinforcement learning agents have been used for working memory tasks ("Generalization of Reinforcement Learners with Working and Episodic Memory", Fortunato et al., 2019), RL agents for visual systems are incredibly brittle to unexpected information (this is classically described in "Vulnerability of Deep Reinforcement Learning to Policy Induction Attacks", A. Behzadan and A. Munir, 2017). As such I am skeptical of RL agents' capacity to perform well in the tasks described when repeated images are subject to transformation, and imagine that focusing on CNNs and recurrent connectivity, which are already common in modeling cognitive function, will be more fruitful. Similarly I have seen that Task 3 of the 5 described tasks in the "Models of working memory" page are limited in behavioral psychophysics data, relative to tasks 1, 2, and 4 (task 5 is a generalization of these) - as such I will struggle to make inferences comparing model behavior and physiology, and so have decided to omit the additional task for sake of computation.

I have not encountered formatting challenges with ImageNet thus far - I have an excess of data that I have restricted, and while I have enough computational resources, limiting runtime has and will be a factor in designing this project.